

```
In [ ]: pip install pandas numpy matplotlib seaborn yfinance statsmodels
```

```
In [2]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import yfinance as yf

from statsmodels.tsa.seasonal import seasonal_decompose

plt.style.use('ggplot')
```

```
In [3]: df = yf.download('AAPL', start='2020-01-01', end='2025-01-01')
```

```
[*****100%*****] 1 of 1 completed
```

```
In [4]: df.head()
```

```
Out[4]:
```

	Price	Close	High	Low	Open	Volume
	Ticker	AAPL	AAPL	AAPL	AAPL	AAPL
	Date					
	2020-01-02	72.333885	72.394093	71.091191	71.344062	135480400
	2020-01-03	71.630638	72.389257	71.406666	71.563205	146322800
	2020-01-06	72.201408	72.239942	70.503546	70.754014	118387200
	2020-01-07	71.861839	72.466322	71.642681	72.211041	108872000
	2020-01-08	73.017830	73.318870	71.565614	71.565614	132079200

```
In [5]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
DatetimeIndex: 1258 entries, 2020-01-02 to 2024-12-31
Data columns (total 5 columns):
#   Column                Non-Null Count  Dtype
---  ---
0   (Close, AAPL)         1258 non-null   float64
1   (High, AAPL)          1258 non-null   float64
2   (Low, AAPL)           1258 non-null   float64
3   (Open, AAPL)          1258 non-null   float64
4   (Volume, AAPL)        1258 non-null   int64
dtypes: float64(4), int64(1)
memory usage: 59.0 KB
```

```
In [6]: df.describe()
```

Out[6]:

	Price	Close	High	Low	Open	Volume
	Ticker	AAPL	AAPL	AAPL	AAPL	AAPL
count		1258.000000	1258.000000	1258.000000	1258.000000	1.258000e+03
mean		151.377785	152.899565	149.680871	151.223469	9.057103e+07
std		41.851999	41.992363	41.629303	41.795805	5.324438e+07
min		54.163700	55.160698	51.324796	55.059309	2.323470e+07
25%		126.279123	127.397093	124.278536	125.724391	5.546825e+07
50%		149.968597	151.709757	148.037172	149.788913	7.628335e+07
75%		175.902020	177.452425	174.337241	175.708231	1.077425e+08
max		257.375580	258.448740	255.994420	256.550862	4.265100e+08

```
In [7]: df.isnull().sum()
```

```
Out[7]: Price    Ticker
        Close    AAPL      0
        High    AAPL      0
        Low     AAPL      0
        Open    AAPL      0
        Volume  AAPL      0
        dtype: int64
```

```
In [8]: df.reset_index(inplace=True)
```

```
In [10]: df.head()
```

```
Out[10]:
```

	Price	Date	Close	High	Low	Open	Volume
	Ticker		AAPL	AAPL	AAPL	AAPL	AAPL
0		2020-01-02	72.333885	72.394093	71.091191	71.344062	135480400
1		2020-01-03	71.630638	72.389257	71.406666	71.563205	146322800
2		2020-01-06	72.201408	72.239942	70.503546	70.754014	118387200
3		2020-01-07	71.861839	72.466322	71.642681	72.211041	108872000
4		2020-01-08	73.017830	73.318870	71.565614	71.565614	132079200

```
In [11]: plt.figure(figsize=(14,6))

plt.plot(df['Date'], df['Close'])

plt.title('Apple Stock Closing Price Trend')
plt.xlabel('Date')
plt.ylabel('Closing Price')

plt.show()
```

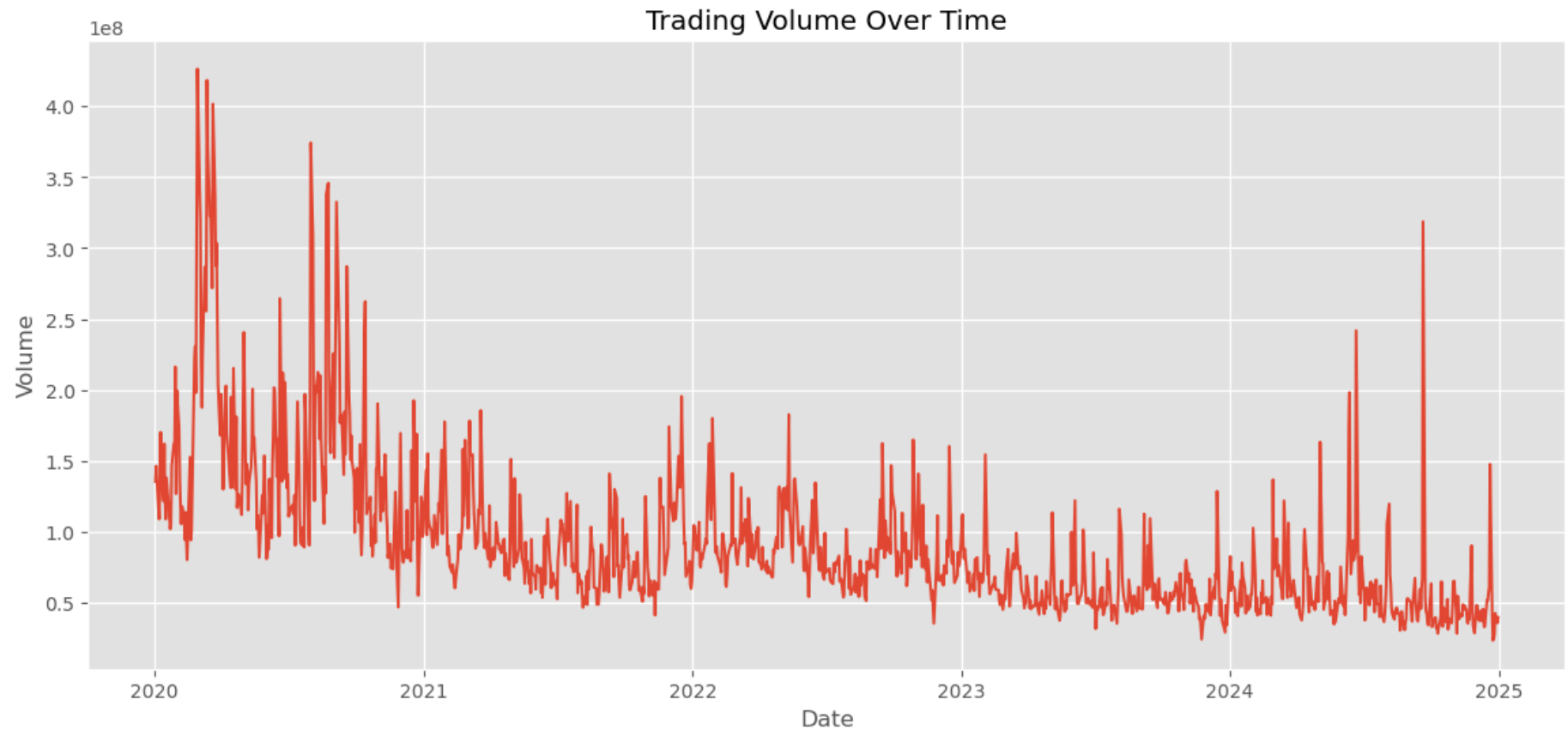


```
In [12]: plt.figure(figsize=(14,6))

plt.plot(df['Date'], df['Volume'])

plt.title('Trading Volume Over Time')
plt.xlabel('Date')
plt.ylabel('Volume')

plt.show()
```



```
In [13]: df['MA30'] = df['Close'].rolling(window=30).mean()
```

```
In [14]: df['MA100'] = df['Close'].rolling(window=100).mean()
```

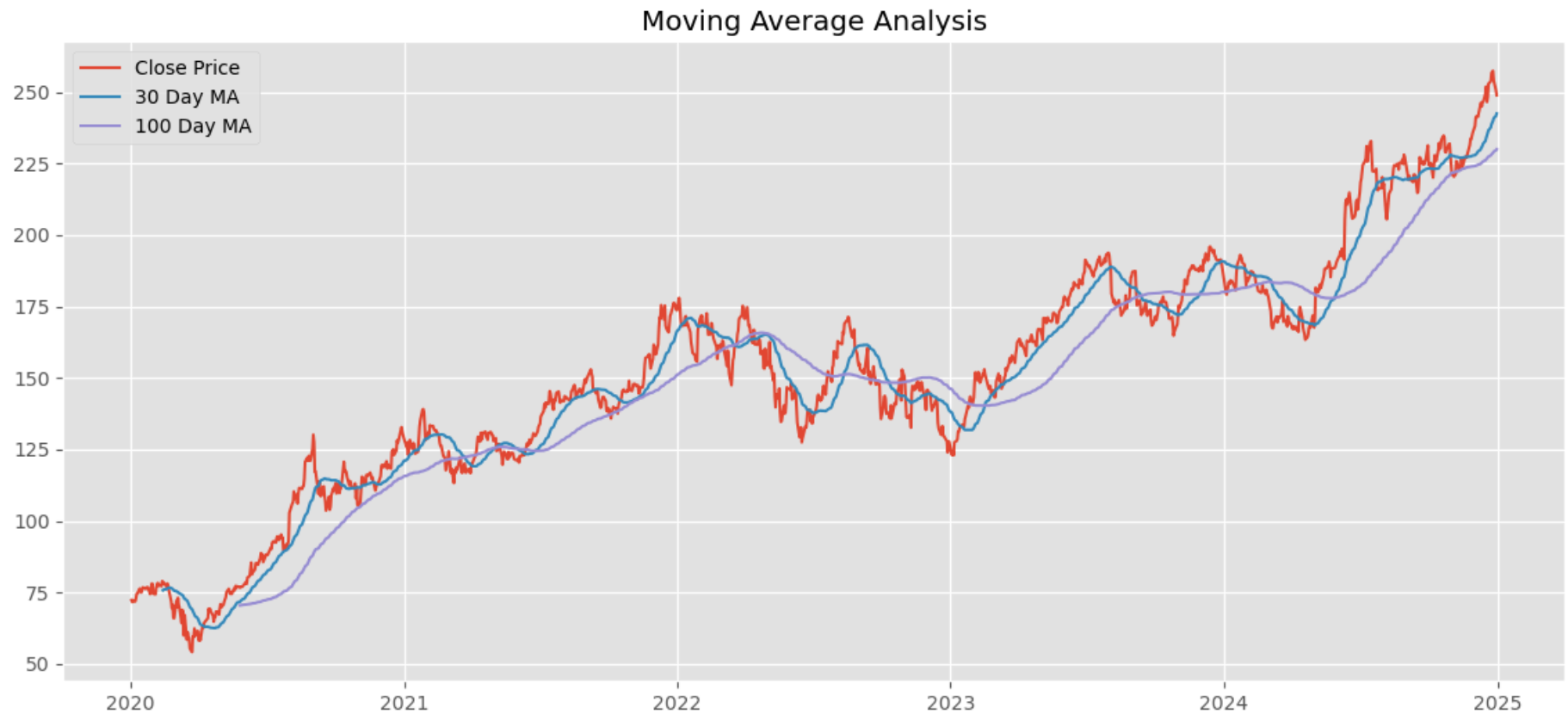
```
In [15]: plt.figure(figsize=(14,6))

plt.plot(df['Date'], df['Close'], label='Close Price')
plt.plot(df['Date'], df['MA30'], label='30 Day MA')
plt.plot(df['Date'], df['MA100'], label='100 Day MA')

plt.legend()

plt.title('Moving Average Analysis')
```

```
plt.show()
```



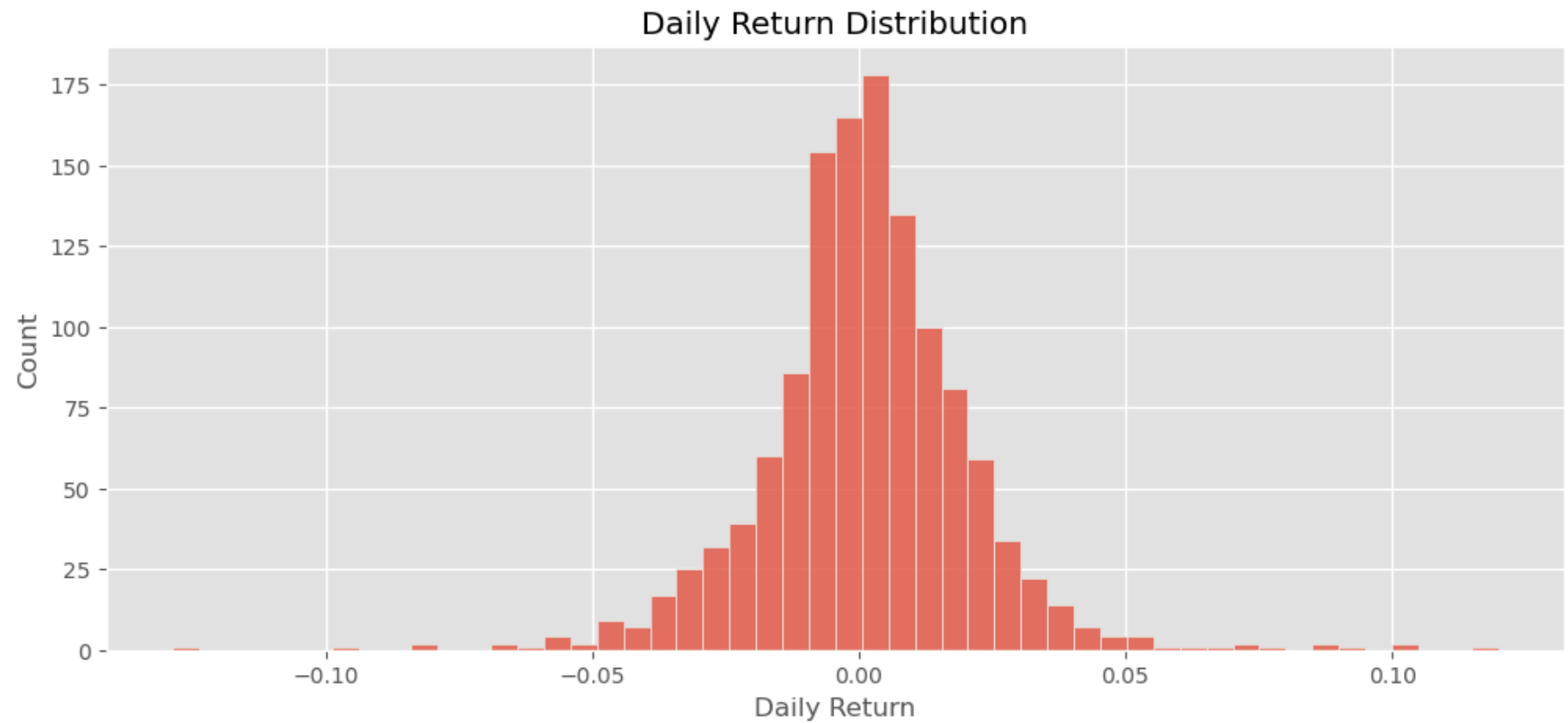
```
In [16]: df['Daily Return'] = df['Close'].pct_change()
```

```
In [17]: plt.figure(figsize=(12,5))

sns.histplot(df['Daily Return'].dropna(), bins=50)

plt.title('Daily Return Distribution')

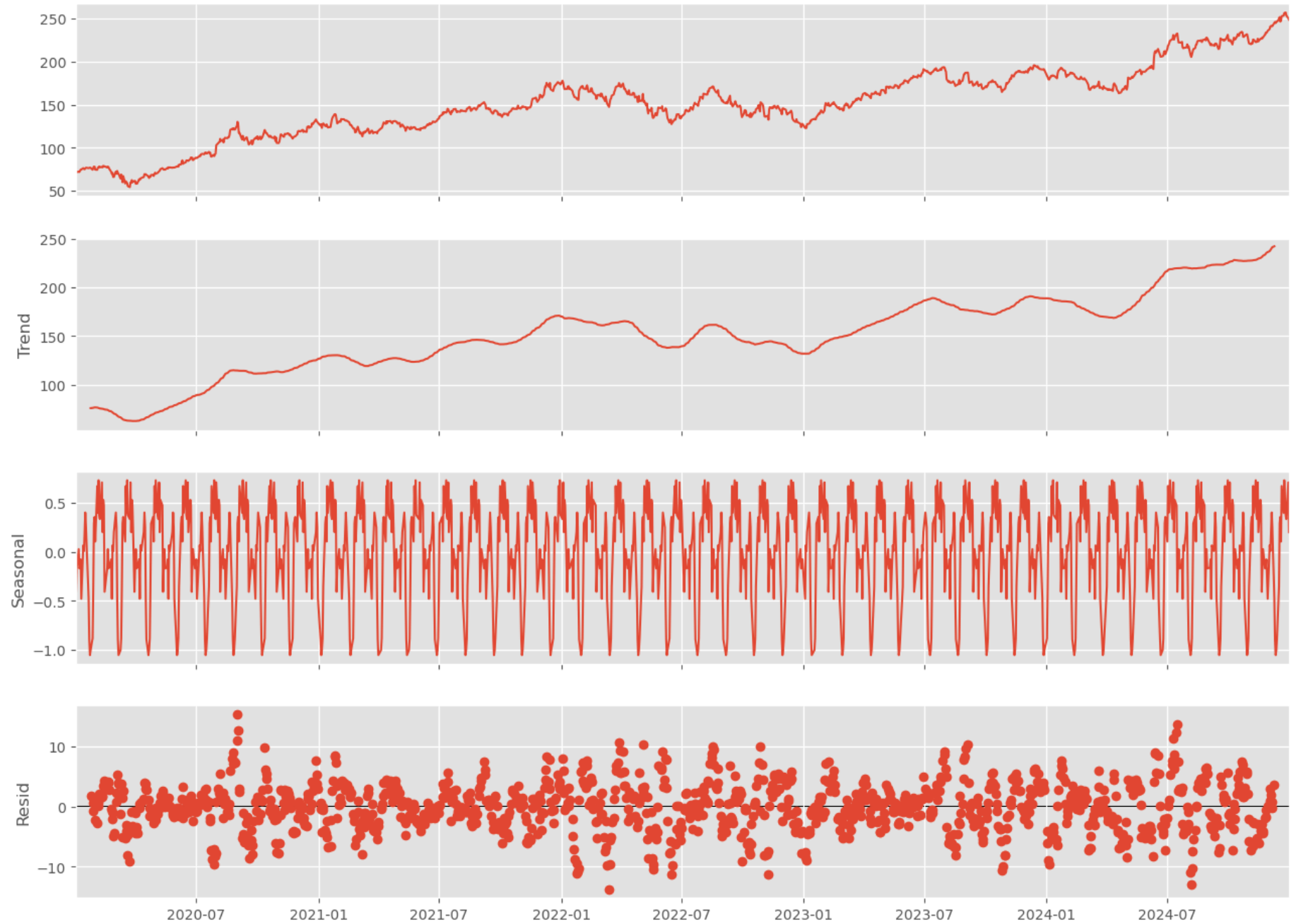
plt.show()
```



```
In [18]: df['Date'] = pd.to_datetime(df['Date'])  
  
df.set_index('Date', inplace=True)
```

```
In [19]: decomposition = seasonal_decompose(df['Close'], model='additive', period=30)
```

```
In [20]: fig = decomposition.plot()  
  
fig.set_size_inches(14,10)  
  
plt.show()
```



```
In [21]: pip install prophet
```

```

Defaulting to user installation because normal site-packages is not writeable
Collecting prophet
  Downloading prophet-1.3.0-py3-none-win_amd64.whl.metadata (3.6 kB)
Collecting cmdstanpy>=1.0.4 (from prophet)
  Downloading cmdstanpy-1.3.0-py3-none-any.whl.metadata (4.2 kB)
Requirement already satisfied: numpy>=1.15.4 in c:\programdata\anaconda3\lib\site-packages (from prophet) (2.3.5)
Requirement already satisfied: matplotlib>=2.0.0 in c:\programdata\anaconda3\lib\site-packages (from prophet) (3.10.6)
Requirement already satisfied: pandas>=1.0.4 in c:\programdata\anaconda3\lib\site-packages (from prophet) (2.3.3)
Collecting holidays<1,>=0.25 (from prophet)
  Downloading holidays-0.97-py3-none-any.whl.metadata (53 kB)
Requirement already satisfied: tqdm>=4.36.1 in c:\programdata\anaconda3\lib\site-packages (from prophet) (4.67.1)
Collecting importlib_resources (from prophet)
  Downloading importlib_resources-7.1.0-py3-none-any.whl.metadata (4.0 kB)
Requirement already satisfied: python-dateutil<3,>=2.9.0.post0 in c:\programdata\anaconda3\lib\site-packages (from holidays<1,>=0.25->prophet) (2.9.0.post0)
Requirement already satisfied: six>=1.5 in c:\programdata\anaconda3\lib\site-packages (from python-dateutil<3,>=2.9.0.post0->holidays<1,>=0.25->prophet) (1.17.0)
Collecting stanio<2.0.0,>=0.4.0 (from cmdstanpy>=1.0.4->prophet)
  Downloading stanio-0.5.1-py3-none-any.whl.metadata (1.6 kB)
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Requirement already satisfied: pytz>=2020.1 in c:\programdata\anaconda3\lib\site-packages (from pandas>=1.0.4->prophet) (2025.2)
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Requirement already satisfied: colorama in c:\programdata\anaconda3\lib\site-packages (from tqdm>=4.36.1->prophet) (0.4.6)
Downloading prophet-1.3.0-py3-none-win_amd64.whl (12.1 MB)
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Downloading holidays-0.97-py3-none-any.whl (1.5 MB)
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Downloading cmdstanpy-1.3.0-py3-none-any.whl (99 kB)
Downloading stanio-0.5.1-py3-none-any.whl (8.1 kB)
Downloading importlib_resources-7.1.0-py3-none-any.whl (37 kB)
Installing collected packages: stanio, importlib_resources, holidays, cmdstanpy, prophet

----- 1/5 [importlib_resources]
----- 2/5 [holidays]
----- 2/5 [holidays]
----- 2/5 [holidays]
----- 2/5 [holidays]
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```

13/17

```

----- 4/5 [prophet]
----- 4/5 [prophet]
----- 4/5 [prophet]
----- 4/5 [prophet]
----- 4/5 [prophet]
----- 5/5 [prophet]

```

Successfully installed cmdstanpy-1.3.0 holidays-0.97 importlib_resources-7.1.0 prophet-1.3.0 stanio-0.5.1

Note: you may need to restart the kernel to use updated packages.

WARNING: The scripts install_cmdstan.exe and install_cxx_toolchain.exe are installed in 'C:\Users\shiva\AppData\Roaming\Python\Python313\Scripts' which is not on PATH.

Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.

```
In [22]: from prophet import Prophet
```

```
In [23]: forecast_df = df.reset_index()[['Date', 'Close']]

forecast_df.columns = ['ds', 'y']
```

```
In [24]: model = Prophet()

model.fit(forecast_df)
```

```

17:48:22 - cmdstanpy - INFO - Chain [1] start processing
17:48:34 - cmdstanpy - INFO - Chain [1] done processing

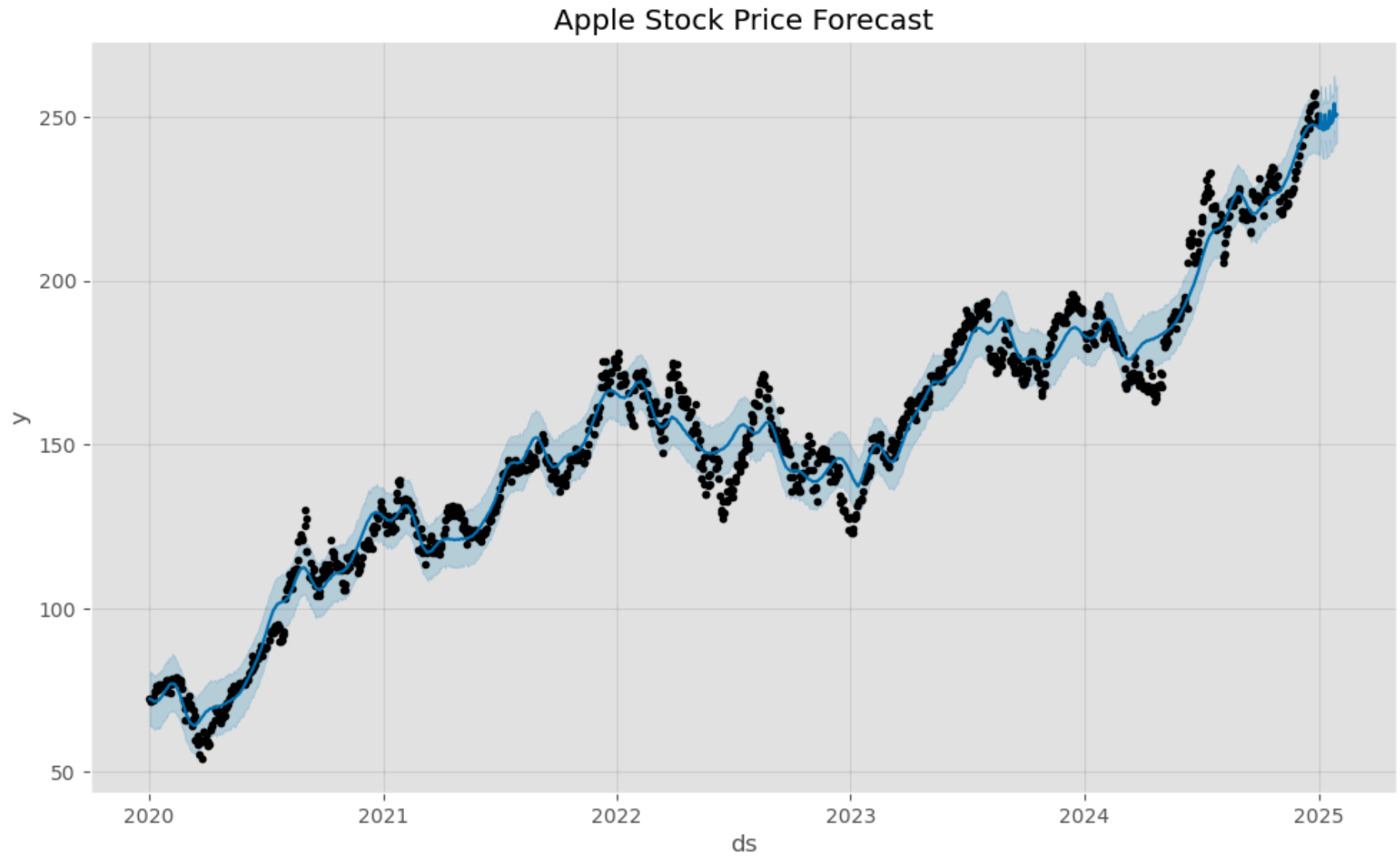
```

```
Out[24]: <prophet.forecaster.Prophet at 0x240b4c9f4d0>
```

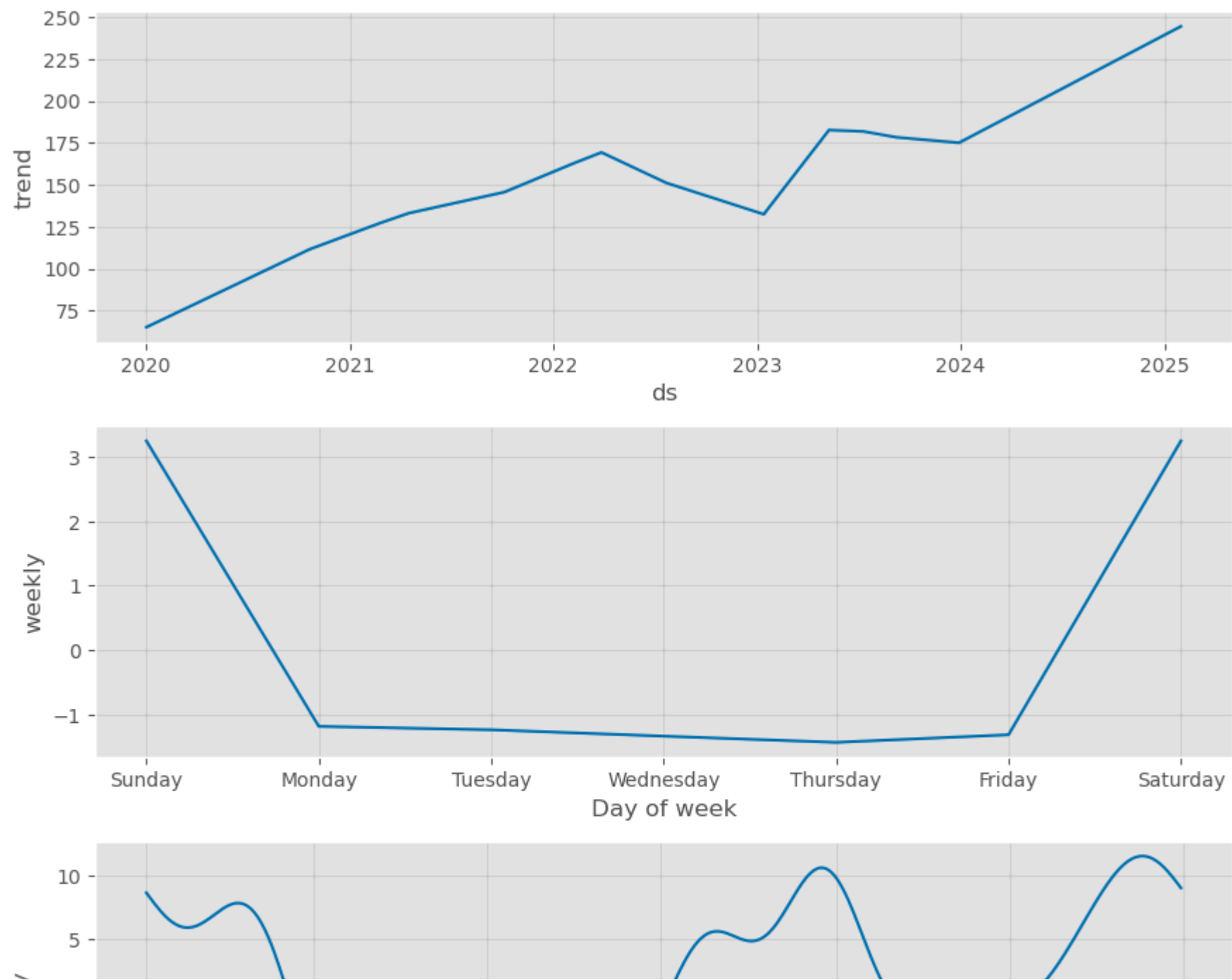
```
In [25]: future = model.make_future_dataframe(periods=30)
```

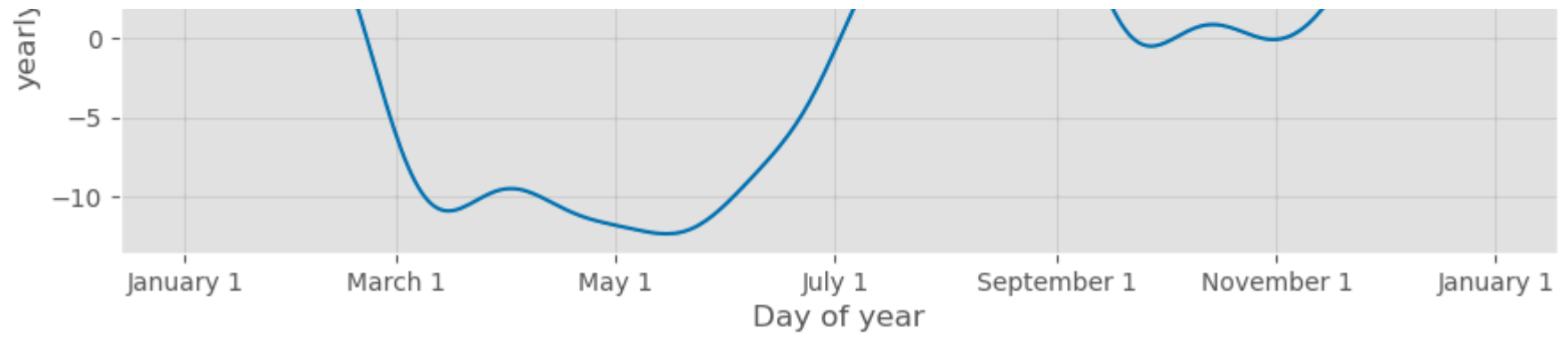
```
In [26]: forecast = model.predict(future)
```

```
In [28]: model.plot(forecast)
plt.title("Apple Stock Price Forecast")
plt.show()
```



```
In [29]: model.plot_components(forecast)
plt.show()
```





In []: